5 Fig. 5.1 shows an aluminium block after leaving a furnace in a factory.

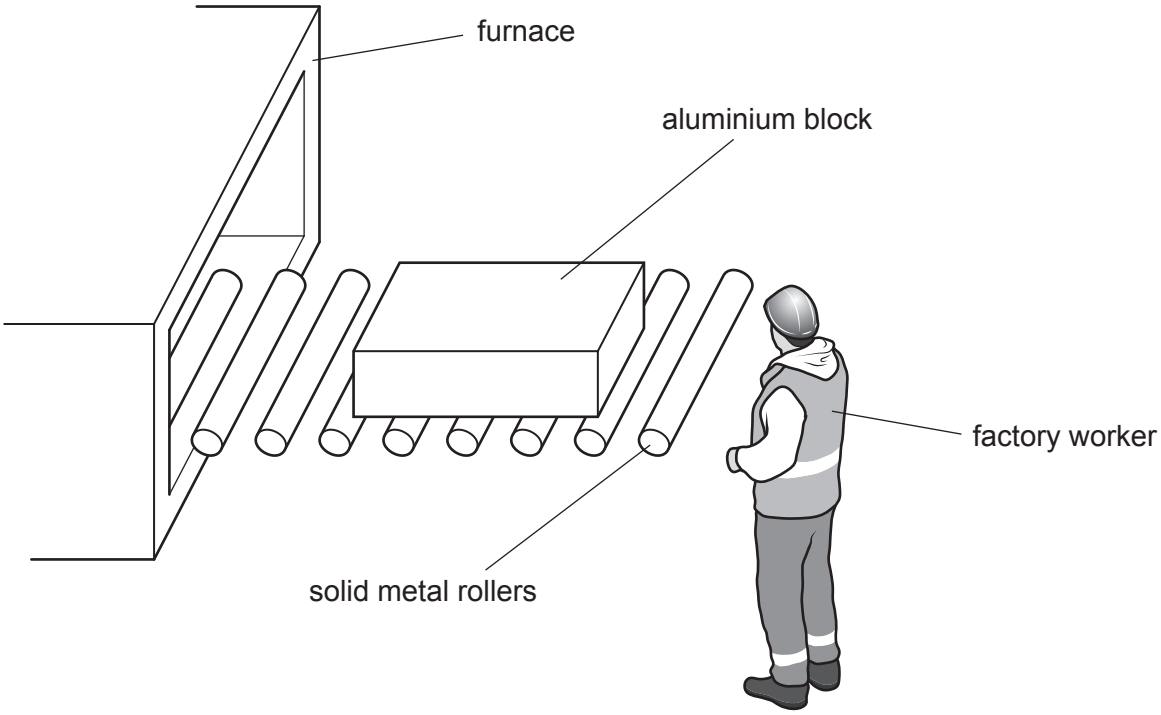


Fig. 5.1

(a) The mass of the block is 1200 kg and it is heated in the furnace from 20 °C to 380 °C. The aluminium block does not melt.
The specific heat capacity of aluminium is 960 J / (kg °C).

Calculate the thermal energy gained by the block in the furnace.

thermal energy = ... [3]

(b) Fig. 5.1 shows a factory worker standing 3 m from the block.

State and explain the main process by which thermal energy is transferred to the worker.

.....

.....

.....

..... [3]

© UCLES 2022

(c) State and explain the main process by which thermal energy is transferred from the outer surface of the solid metal rollers to their interior.

.....

.....

.....

..... [3]

[Total: 9]